

Installing IBM Data Cataloging

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IBM Data Cataloging service is a modern metadata management software that provides data insight for exabyte-scale heterogeneous file, object, backed storage costs, uncovering hidden data value, and reducing the risk of massive data stores.

Before you begin

- Meet the system requirements to install the IBM Data Cataloging service.
- The following details are a base line for finding the resources that are needed for IBM Fusion IBM Data Cataloging service deployment. Based on the following are the resource values that are calculated per compute node: You must have at least two worker nodes, each with the same amount of resources.
- IBM Fusion IBM Data Cataloging service must have dedicated compute resources. Make sure that you have enough to cover the resources limits to prevent resource contention.

Table 1. Profile requirements

CPU	RAM	Disk space	Network
77	162 GB	120 GB	10 GB

- The standard deployment for IBM Data Cataloging service project requests and limits. Ensure you set the resource quota values for IBM Data Cataloging service.

```
kind: ResourceQuota
apiVersion: v1
metadata:
  name: dcs-resource-quota
  namespace: ibm-data-cataloging
spec:
  hard:
    limits.cpu: '72'
    limits.memory: 115Gi
    pods: '120'
    requests.cpu: '18'
    requests.memory: 26Gi
```

 **Important:** For the IBM Data Cataloging service to run successfully on all platforms, ensure that the storage classes have the following attributes:

- ReadWriteMany (RWX) permissions
- volumeBindingMode set to Immediate
- AllowVolumeExpansion set to true

- If you have not configured the IBM operator catalog, then configure it. For the procedure to add IBM operator catalog, see [Adding the IBM operator catalog](#).
- Go through troubleshooting information related to the installation of IBM Data Cataloging. See [IBM Data Cataloging service installation and upgrade issues](#).
- IBM Data Cataloging can be deployed as containers on Red Hat OpenShift Virtualization Engine (OVE) but must run on normal worker nodes.

Procedure

1. Go to Services page in IBM Fusion user interface.
2. In the **Available** section, click **Data cataloging** tile.
3. In the Data cataloging window, go through the details of the service and click **Install**.
4. In the Install service message box, select a **Storage class**.

 **Important:**

- If you want to use Global Data Platform as the storage provider, then it is recommended to select the default storage class `ibm-spectrum-storage` storage class. You can also use a custom storage class that matches the requirements.
- For deployments operating in FIPS mode, you must specify or select a storage class that uses encrypted storage.
 - If the default storage classes (`ibm-spectrum-fusion` or `ocs-storagecluster-cephfs`) are FIPS-compliant and use **encrypted** storage.
 - If not, you must configure and select a **custom encrypted storage class** that meets FIPS requirements.
 - For storage based on node SSDs, ensure the SSDs are encrypted at deployment time using encryption methods that meets FIPS requirements.
 - For storage based on Red Hat® OpenShift® Data Foundation, configure encrypted Red Hat OpenShift Data Foundation storage and use a custom storage class.

5. Click **Install**. In case of failures, go through the downloaded logs to understand the cause of the failure and fix the issue. For more information about troubleshooting, see [Troubleshooting IBM Data Cataloging](#).
6. Validate the installation.
 - IBM Fusion user interface:

After you enable the IBM Data Cataloging service, you can view the service version and health status. From the ellipsis menu, you can download I disappears automatically after some time.

Table 2. Health states IBM Data Cataloging service

State	Description
Installing	Service installation is in progress
Upgrading	Service upgrade is in progress
Healthy	Service is healthy
Degraded	Service is not healthy

- Verify the IBM Data Cataloging installation from the OpenShift Container Platform web console:
 - Run the following command to verify the installed operators.

```
oc -n ibm-data-cataloging get csv
```

Expected output

NAME	DISPLAY	VERSION
amqstreams.v2.3.0-1	Red Hat Integration - AMQ Streams	2.3.0-1
db2u-operator.v3.1.0	IBM Db2	3.1.0
ibm-spectrum-discover-operator.v210.0.0	IBM Storage Discover	210.0.0-1678324'

- Run the following command to verify IBM Data Cataloging status.

```
oc -n ibm-data-cataloging get isd
```

Expected output

NAME	READY	SUMMARY	ERROR	AGE	HEALTHSTATUS
data-cataloging-service-instance	True	Awaiting next reconciliation		9d	Healthy

- Run the following command to verify that all Persistent Volume Claims are bounded.

```
oc -n ibm-data-cataloging get pvc
```

Expected output

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES
activelogs-c-isd-db2u-0	Bound	pvc-7e98eead-9a38-48b6-a880-16d730120796	150Gi	RWO
activelogs-c-isd-db2u-1	Bound	pvc-7af71786-1529-4b3c-bbd1-b5d1ee733d33	150Gi	RWO
c-isd-meta	Bound	pvc-1ba20699-ddb8-4600-bf9f-4d637b14da49	50Gi	RWX
data-c-isd-db2u-0	Bound	pvc-488540db-c951-4f48-b65c-814bf634ca33	300Gi	RWO
data-c-isd-db2u-1	Bound	pvc-56a3105a-347e-4b55-b11a-d0a1d351d622	300Gi	RWO
data-isd-sasl-kafka-0	Bound	pvc-3924c9cc-173d-4244-90fa-a234325dc04b	100Gi	RWO
data-isd-sasl-zookeeper-0	Bound	pvc-3c66397b-ac03-4ed8-841a-e33d83355a23	64Gi	RWO
data-isd-ssl-kafka-0	Bound	pvc-ce0ebeaf-d2b6-4e40-8c31-e31d45a208e5	100Gi	RWO
data-isd-ssl-zookeeper-0	Bound	pvc-cd3fdd20-343b-49ca-ad1a-690f8eec3e60	64Gi	RWO
isd-backup	Bound	pvc-2e077c8e-e992-4e1a-8b98-d24bfdbb5343	250Gi	RWX
isd-data	Bound	pvc-b98e124f-1f98-43e5-9bf6-c8758c681192	250Gi	RWX
tempts-c-isd-db2u-0	Bound	pvc-7fdc9ac5-9158-48ac-bad3-3c24590f1f2b	50Gi	RWO
tempts-c-isd-db2u-1	Bound	pvc-a2adb668-a362-4f61-8ed3-ae103f9b2cd3	50Gi	RWO

- Run the following command to verify Db2u cluster detailed progress.

```
oc -n ibm-data-cataloging get formations.db2u.databases.ibm.com isd -o go-template='{{range .status.components}}{{printf "%s,%s,%s\}}
```

Expected output

account	account-ibm-data-cataloging-isd	OK
PersistentVolumeClaim	c-isd-meta	OK
secret	c-isd-sshkeys-db2uhausr	OK
secret	c-isd-sshkeys-db2uadm	OK

secret	c-isd-sshkeys-db2instusr	OK
secret	c-isd-ldappassword	OK
secret	c-isd-ldapblueadminpassword	OK
secret	c-isd-instancepassword	OK
secret	c-isd-db2u-lic	OK
secret	c-isd-certs-wv-rest	OK
secret	c-isd-certs-db2u-api	OK
configmap	c-isd-db2uconfig	OK
configmap	c-isd-db2regconfig	OK
configmap	c-isd-db2dbmconfig	OK
configmap	c-isd-db2dbconfig	OK
service	c-isd-ldap	OK
Deployment	c-isd-ldap	OK
service	c-isd-tools	OK
Deployment	c-isd-tools	OK
configmap	c-isd-db2u-api	OK
service	c-isd-db2u-engn-svc	OK
service	c-isd-db2u-head-engn-svc	OK
Job	c-isd-instdb	OK
service	c-isd-etcd	OK
StatefulSet	c-isd-etcd	OK
service	c-isd-db2u-rest-svc	OK
networkpolicy	c-isd-rest-ext	OK
Deployment	c-isd-rest	OK
service	c-isd-db2u-graph-svc	OK
networkpolicy	c-isd-graph-ext	OK
Deployment	c-isd-graph	OK
service	c-isd-db2u-internal	OK
service	c-isd-db2u	OK
StatefulSet	c-isd-db2u	OK
networkpolicy	c-isd	OK
networkpolicy	c-isd-ext	OK
Job	c-isd-restore-morph	OK

- e. Run the following command to verify the Db2u cluster CR status.

```
oc -n ibm-data-cataloging get db2ucluster
```

Expected output

NAME	STATE	MAINTENANCESTATE	AGE
isd	Ready	None	9d

- f. Run the following command to verify that the ports 50000 and 50001 are listing.

```
oc exec -n ibm-data-cataloging -it $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector name=dashmpp-head-
```

Expected output

Defaulted container "db2u" out of: db2u, init-labels (init), init-kernel (init)

(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)

Active Internet connections (only servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	PID/Program name
tcp	0	0	10.254.4.11:60000	0.0.0.0:*	LISTEN	23707/db2sysc 0
tcp	0	0	10.254.4.11:60001	0.0.0.0:*	LISTEN	23712/db2sysc 1
tcp	0	0	0.0.0.0:9443	0.0.0.0:*	LISTEN	-
tcp	0	0	0.0.0.0:50022	0.0.0.0:*	LISTEN	-
tcp	0	0	0.0.0.0:50000	0.0.0.0:*	LISTEN	23707/db2sysc 0
tcp	0	0	0.0.0.0:50001	0.0.0.0:*	LISTEN	23707/db2sysc 0
tcp6	0	0	:::50052	:::*	LISTEN	-
tcp6	0	0	:::50022	:::*	LISTEN	-

- g. Run the following command to verify that workload is PUREDATA_OLAP.

```
oc exec -n ibm-data-cataloging -c db2u -it $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector name=dashn
```

Expected output
DB2_WORKLOAD=PUEDATA_OLAP

- h. Run the following command to verify the database encryption configurations.

```
oc -n ibm-data-cataloging exec -c db2u $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector name=dashmpp-f
```

Encrypted database = YES

- i. Run the following command to verify the Kafka CRs readiness.

```
oc -n ibm-data-cataloging get kafka
```

Expected output

NAME	DESIRED KAFKA REPLICAS	DESIRED ZK REPLICAS	READY	WARNINGS
isd-sasl	1	1	True	
isd-ssl	1	1	True	

- j. Run the following command to verify that the IBM® Spectrum Discover pods are running.

```
oc -n ibm-data-cataloging get pod -l component=discover
```

Expected output

NAME	READY	STATUS	RESTARTS	AGE
isd-api-7564475d98-cs6nx	1/1	Running	0	9d
isd-auth-5fcb8bd95f-jvzwp	1/1	Running	0	9d
isd-backup-restore-689fc64f7-pgx22	1/1	Running	0	9d
isd-connmgr-7d4cdb4cc7-xnkzm	1/1	Running	1 (9d ago)	9d
isd-consumer-ceph-le-5f9b5d4676-4fsgg	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-9k9cs	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-bphqp	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-cc2pz	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-chlvk	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-fchp5	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-jlbrw	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-l78cc	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-lpcsn	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-n2wqt	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-54qm9	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-6hsq9	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-9p4p5	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-dhlgc	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-hr76h	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-mkvx8	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-tvscz	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-vrrd4	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-w59c2	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-xqnc1	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-4klcc	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-8x6w7	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-fqs5m	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-fw8p	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-h78g5	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-lvw15	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-nhzn	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-q2l8b	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-z6tfj	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-zv2zw	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-5mrrg	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-8rphj	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-b5dbv	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-b5lv2	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-d9gbm	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-f5rxf	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-m8nw6	1/1	Running	0	9d

isd-consumer-file-scan-5c7565bf7b-ntfbb	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-ql55x	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-qvggm	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-6s75x	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-845bx	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-8kmnf	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-gntbt	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-mlnfb	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-p9lpt	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-rckm8	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-vkmpd	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-vmh8x	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-zm4gf	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-6dw9h	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-9nhsz	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-dnwpw	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-dz2mf	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-h2ffj	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-hdrxg	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-nf779	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-pf6n4	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-q8gqf	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-xzmvh	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-49vkb	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-54pb8	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-5sr89	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-98lm9	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-d4zlj	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-fc82j	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-mnsn2	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-r2kc6	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-tglisz	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-zbhgv	1/1	Running	0	9d
isd-contentsearchagent-5d7bfbd887-r487x	1/1	Running	1 (9d ago)	9d
isd-db-schema-xm5cn	0/1	Completed	0	9d
isd-db2whrest-6c6bbfcbcb6-9v6v2	1/1	Running	1 (9d ago)	9d
isd-generate-license-6g94v	0/1	Completed	0	9d
isd-importtags-64498dbdd8-5kqpb	1/1	Running	1 (9d ago)	9d
isd-keystone-7dc84b69f5-sw6nn	1/1	Running	0	9d
isd-license-check-27982200-c9gkm	0/1	Completed	0	2d21h
isd-license-check-27983640-ldxv2	0/1	Completed	0	45h
isd-license-check-27985080-rhzjl	0/1	Completed	0	21h
isd-policyengine-598f565867-zg5sn	1/1	Running	1 (9d ago)	9d
isd-producer-ceph-le-76b9885854-557b7	1/1	Running	0	9d
isd-producer-cos-le-85d446c9d8-zvzmg	1/1	Running	0	9d
isd-producer-cos-scan-d584f646d-2t669	1/1	Running	0	9d
isd-producer-file-scan-5586b48fd-zkt7q	1/1	Running	0	9d
isd-producer-protect-scan-7b66bb7c64-wz6qq	1/1	Running	0	9d
isd-producer-scale-le-8d7bc64b4-nvqbl	1/1	Running	0	9d
isd-producer-scale-scan-5df647956-qfl6p	1/1	Running	0	9d
isd-proxy-5f4d7dd8ff-44fwt	1/1	Running	0	9d
isd-reports-c6kmq	0/1	Completed	0	9d
isd-scaleafmdatamover-85cc44b566-644vx	1/1	Running	1 (9d ago)	9d
isd-scaleilmdatamover-6b86dd7b58-j28sn	1/1	Running	1 (9d ago)	9d
isd-sdmonitor-6ffffc999b7-s2srd	1/1	Running	0	9d
isd-tikaserver-79548bdb4-ssft5	1/1	Running	0	9d
isd-ui-backend-f6c4b65d7-dwzrs	1/1	Running	0	9d
isd-ui-frontend-9659fc847-vcbmvm	1/1	Running	0	9d

k. Run the following command to verify the console route availability.

```
oc -n ibm-data-cataloging get route console
```

Expected output

NAME	HOST/PORT	PATH	SERVICES	PORT	TERMINATION	WILDCARD
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console console-ibm-data-cataloging.apps.ocp2.vmlocal isd-svc <all> reencrypt None

Parent topic:

→ [Data Cataloging](#)